



Exposure and health risks of livestock air resistomes

Wei Sun^{a,b,c,1}, Jun Wang^{a,b,c,1}, Guangdong Wang^{a,b,c} , Lan Jiang^{a,b,c}, Wenlu Feng^{a,b,c}, Shuangsoo Dang^d, Mei Li^d, Shuo Jiao^b , Gehong Wei^e, Jie Gu^{a,b,c,2} , James M. Tiedje^{a,f,2} , and Xun Qian^{a,b,c,2}

Affiliations are included on p. 9.

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Most of the global antibiotic consumption is by the livestock industry, making livestock farms a hotspot of antibiotic resistance genes (ARGs). Farm air poses direct ARG exposure to workers, but the health risks of air resistomes remain unclear. We evaluated the human exposure and health risks of air resistomes in pig and chicken farms and compared air resistomes in Chinese farms to those in European farms given their long-term restrictions on use of antibiotics in livestock. We found that livestock air was highly enriched in ARGs, with each cell harboring seven times more ARGs than urban air. The daily ARG inhalation of farm workers was equivalent to several years of ARG inhalation by urban residents. ARGs encoding resistance to last-resort antibiotics such as *mcr-1* and *tetX* were detected in farm air, and *tetX* variants were prevalent in both Chinese and European farms. ARGs in livestock air were highly associated with mobile genetic elements, and conjugation experiments confirmed their cross-phyla transferability. The projected resistome risk of farm air was significantly higher than well-recognized ARG hotspots like air from hospitals, sewage treatment plants, and from animal manures. The diversity, abundance, and risk score of air resistomes in Chinese farms were significantly higher than those in European farms, suggesting that long-term restriction of antibiotic use mitigates antibiotic resistance in the livestock environment. Our results underscore the high exposure of farm workers to ARGs via farm air and highlight its role in ARG dissemination, supporting the importance of antibiotic stewardship practices in combating antibiotic resistance.

antibiotic resistance | livestock farm | health risk | airborne | antibiotic stewardship

The rapid development and transmission of antibiotic resistance genes (ARGs) is a major crisis to global public health. Approximately 4.95 million people die of antibiotic resistant related diseases every year, and the crisis is continuously worsening (1). Growing evidence suggests that transmission of ARGs from environment to clinical settings is an important reason for the increased spread of antibiotic resistance (2–4). Livestock farming consumed over 70% of global antibiotics (5), plus the coexistence of pathogens and human commensals, make livestock environments a special concern (6, 7).

Chemical and biological pollutants carried by air are easily transported and spread at local and regional scales (8). Compared to environmental matrices such as soils and waters, airborne antibiotic resistant bacteria (ARB) and ARGs more directly and frequently lead to human exposure. Recent studies have shown that aerosols and air particles in wastewater plants (9), public transportation environment (10), and haze (11, 12) are enriched ARG reservoirs and play important roles in ARG dissemination. In addition to the above environments, intensive animal husbandry and their feces in a relatively enclosed space result in high abundance of bacteria and ARGs in livestock farm air (13, 14). Although inhalation exposure may not be the only reason, a significant increase of ARG abundance in the intestinal microbiome of farm workers and veterinary students has been reported (15, 16).

The occurrence of ARGs in the air of livestock farms has been surveyed in several studies (17, 18), but the comprehensive health risk of antibiotic resistomes mediated by livestock farm air remains unclear. Host range and gene mobility are two essential factors in evaluating health risks of ARGs (19). Metagenomics and bioinformatic analyses provide opportunities for assessing these traits in environmental samples (20, 21). However, metagenomic assembly has poor performance in assembling low-abundance genes and genes flanked by repeat sequences, restricting the risk evaluation of ARGs (22). In addition, the predicted mobility of ARGs may not correspond to the actual horizontal transfer ability, so it is essential to use complementary approaches such as culture-based and conjugation methods to validate ARG transfer to environmental and clinical species (2).

In this study, we sampled total suspended particles (TSP) from pig farms, chicken farms, near-farm residential areas, and urban areas and evaluated exposure and health risks of air resistomes by integrating information from metagenomics, ARB isolation, and conjugation

Significance

Livestock farm air is a potential transmission route for antibiotic resistance to humans, yet its health risks are not suitably evaluated. We demonstrate that livestock air is a significant antibiotic resistance gene (ARG) exposure pathway for farm workers and surrounding residents, and that farm air ARGs are highly transferrable to human pathogens. To understand the impacts of antibiotic stewardship on ARG health risks, we used European farm air as a benchmark and compared it with Chinese farms recently implemented restrictions on animal antibiotic use. The longer-term restriction of antibiotic use in animal farming corresponded to a reduction in antibiotic resistances and its risk score, highlighting that restricting use is an important approach in mitigating risks for the One Health initiative.

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¹W.S. and J.W. contributed equally to this work.

²To whom correspondence may be addressed. Email: gujie205@sina.com, tiedje@msu.edu, or qianxun@nwafu.edu.cn.

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experiments. European countries have the longest history in restricting antibiotic use in animal farming, while other countries like China are enforcing similar policies in recent years (23, 24). Understanding the impacts of antibiotic stewardship on health risks of ARGs is critical for policymaking to curb antibiotic resistance. Hence, we used metagenomes from local livestock air samples and from public databases to compare the profile and projected health risks of air resistomes in livestock farms between China and nine European countries. Our objectives were to 1) clarify the profile, host range, and potential mobility of air resistomes in livestock farms; 2) evaluate the exposure, conjugative transferability, and health risks of farm air resistomes; and 3) compare the profile and health risk of air resistomes in Chinese livestock farms to those in European livestock farms. This study provides comprehensive insights into the health risks of ARGs mediated by farm air and highlights the role of antibiotic stewardship in mitigating antibiotic resistance.

Results

Livestock Farm Air Is Highly Enriched in ARGs. The diversity and abundance of ARGs in the air from Chinese and European livestock farms were significantly higher than those in urban air (Fig. 1A and SI Appendix, Fig. S1). On average, farm air (157 ± 37)

detected 41 more ARG subtypes than urban air (114 ± 50). The air resistome abundance in and nearby farms were over 4.6 times of that in urban area. We detected 38 and 33 identical ARG sequences in the chicken and pig farms with their corresponding nearby air samples respectively (SI Appendix, Fig. S2). The profiles of air resistomes were significantly (permanova, $P < 0.01$) affected by animal type, country, and air type (farm vs urban air) (Fig. 1B). The highest resistome abundance was observed in pig and chicken farms in China (5.2 ± 1.3 copies per cell), which was eight times of that in urban air samples (0.68 ± 0.36 copies per cell) (Fig. 1A). Chinese farms (211 ± 28 subtypes) had over 60 more ARG subtypes than European farms (SI Appendix, Fig. S1). The resistome abundance in Chinese chicken and pig farms was 70% and 100% higher than those in European chicken and pig farms, respectively (Fig. 1A). Aminoglycoside and chloramphenicol resistance genes were the most enriched types in Chinese farms, and they were over 200% higher in Chinese farms than European farms (SI Appendix, Fig. S3). The abundance of multidrug resistance genes in the air of European pig farms was significantly lower than that in all other farms ($P < 0.01$), and it was only 21 to 27% of that in other farms.

Compared to urban air, 50 to 55 and 25 to 27 ARGs were enriched by more than 10 times in Chinese and European farms, respectively (Fig. 1C). Among them, *aph(2'')-II*, *vatB*, *vanG*, and

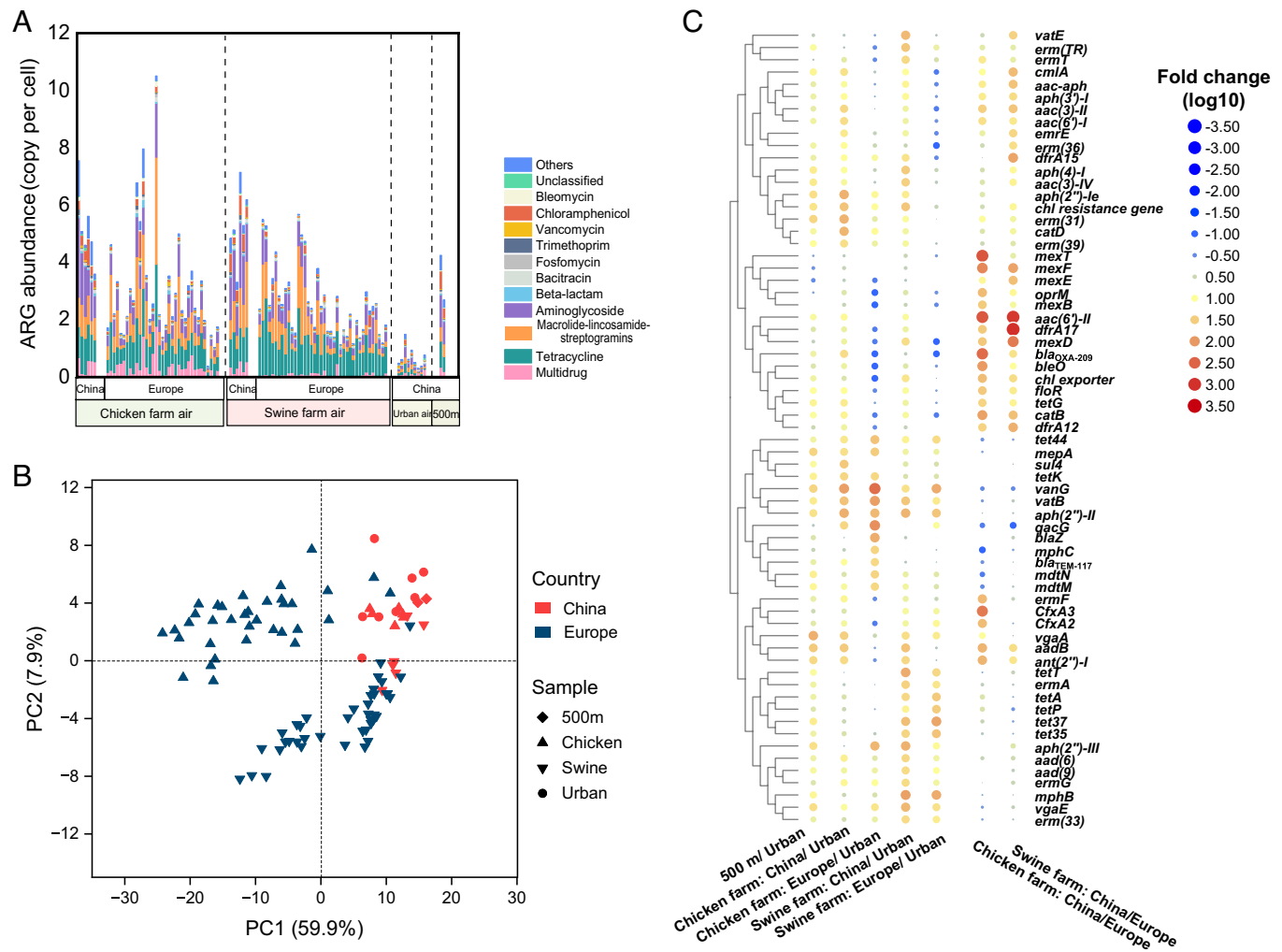


Fig. 1. ARGs in the air of Chinese and European livestock farms and urban regions. (A) Relative abundance of ARGs. (B) PCA showing dissimilarities of resistome profile among different air samples. (C) Heatmap showing ARG subtypes significantly enriched in livestock farm air (compared to urban air) and in Chinese farms (compared to European farms).

tet44 were enriched over 10 times in all farm air samples. Twenty-one ARGs were enriched by over 10 times in Chinese farm air compared to the same type of European farms. *aac(6)-II* was the most enriched (414 to 840 times) ARG in Chinese farms, and *dfrA17* was the most enriched (1,039 times) ARG in Chinese pig farms. In contrast, European farm air enriched in *qacG*, *mphC*, *vanG*, and *tet44*, compared to Chinese farms, and *qacG* was the most enriched ARG (4.7 to 10.1 times).

Source Tracking of Air Microbiome and Resistome in Livestock Farms. There were significant differences in microbial profile between urban air and farm air (SI Appendix, Fig. S4). Compared to urban air, farm air detected higher relative abundance of Bacillota but lower abundance of Pseudomonadota and Cyanobacteriota (SI Appendix, Fig. S5). The top 20 genera accounted for 69.2% of the total bacterial abundance in farm air, while they only accounted for 18.3% in urban air (Fig. 2A). No significant correlation was observed between ARG diversity and microbial diversity in both livestock, but the correlation in urban air was stronger than that in farm air (Fig. 2B). Unlike ARG diversity, the bacterial richness index was significantly higher in urban air than in farm air ($P < 0.001$). Procrustes analysis showed that the resistome profile was significantly correlated with the bacterial community composition ($M^2 = 0.183$, $r = 0.904$, $P < 0.001$) (Fig. 2C).

To understand the contributing sources to microbiome and resistome in farm air, we performed source tracking analysis by collecting and using microbiomes and resistomes from diverse ecosystems as potential sources. The microbial source tracking analysis predicted that the microbiomes in farm air mainly originated from natural air, accounting for over 50% of total microbiome. Animal and human associated sources accounted for 7.8 to 8.6% of air microbiome (Fig. 2D), and 33 to 40% of farm air microbiome was not assigned to a known source. The resistome source tracking suggested that natural air (54.7%) was the dominant source of ARGs in pig farm air, followed by animal-associated environments (37.2%). In contrast, animal-associated environment (55.9%) contributed the most to air resistome in chicken farms, and natural air (25.3%) was the second biggest contributing source.

Many ARGs in Livestock Farm Air Were Carried by Human Pathogens. A total of 68 antibiotic resistant isolates, classified to 17 genera and 40 species, were isolated from air samples in Chinese chicken and pig farms. About 82% of them were resistant to at least three different classes of antibiotics (SI Appendix, Table S1). The isolates in total carried 130 ARG subtypes potentially encoding resistance to 16 antibiotic classes (Fig. 3A; detailed lists in Dataset S1). Twenty-three isolates simultaneously carried

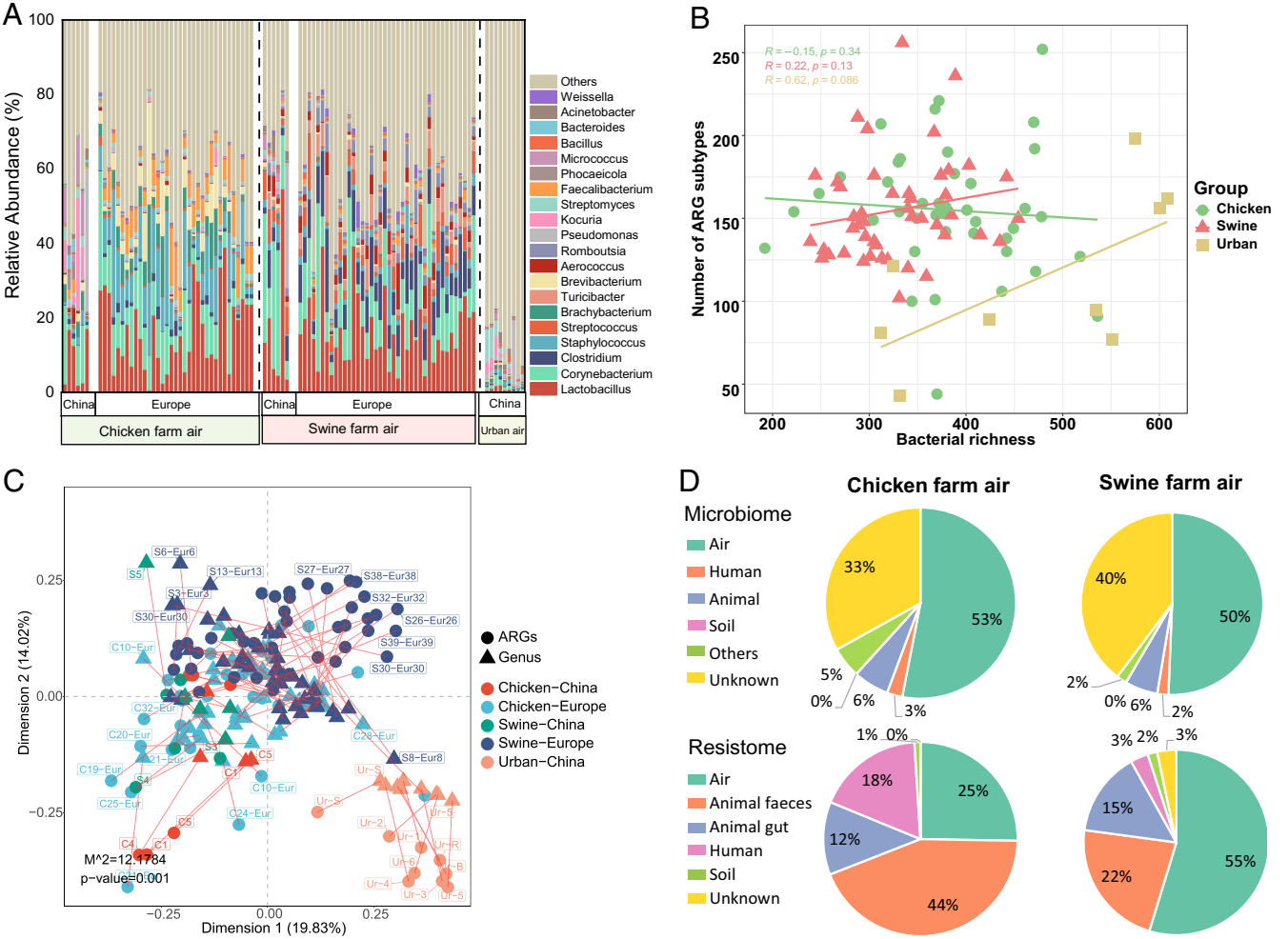


Fig. 2. The microbial community and source composition of air microbiome and resistome in livestock farms. (A) Abundance of top 20 abundant genera in livestock farm air and urban air. (B) Spearman correlation between bacterial richness and ARG diversity in farm air and urban air. (C) Procrustes analysis showing the relationship between ARG profile and bacterial profile (genus level) in livestock farm air and urban air. (D) Source tracking of air microbiome and resistome in livestock farms.

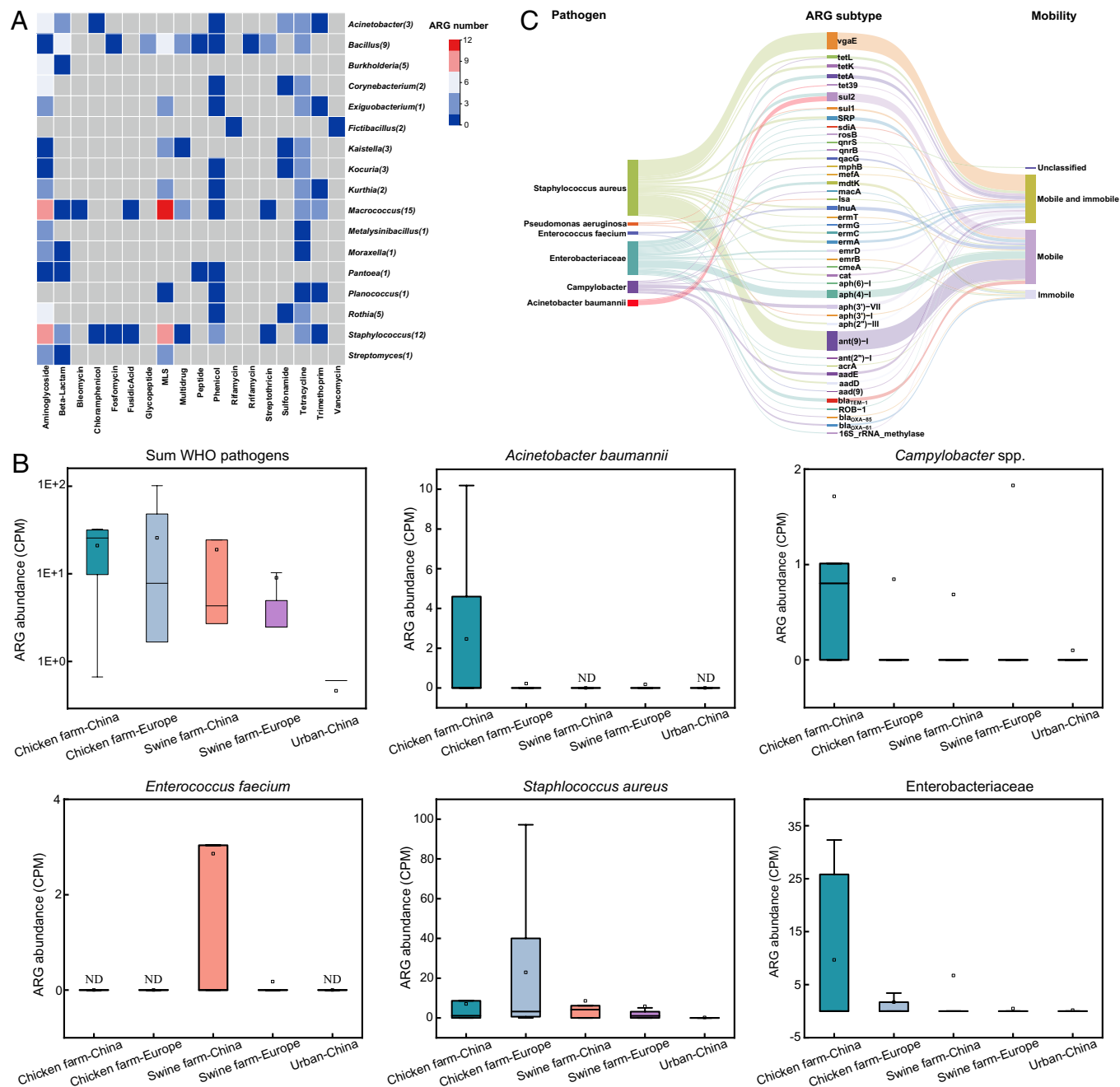


Fig. 3. ARG hosts in the air of livestock farms. (A) Heatmap showing the number of ARGs carried by antibiotic resistant isolates from air samples in Chinese chicken and pig farms. (B) Abundance of ARGs carried by WHO antibiotic-resistant "priority pathogens" in different types of air. (C) Sankey plot showing ARG subtypes and their mobility carried by WHO antibiotic-resistant priority pathogens in livestock air samples.

aminoglycoside, macrolide-lincosamide-streptogramin (MLS), and tetracycline resistance genes. Thirty-four isolates harbored at least one ARG associated with plasmid or ICE (integrative and conjugative element). *Macrocooccus* (15/68) and *Staphylococcus* (12/68) were the main genera of antibiotic resistant isolates; they on average harbored 8.1 ARG subtypes.

The abundance of ARGs carried by the five WHO priority pathogens with antibiotic resistance in Chinese farm air was 17 times of that in urban air and was 50% higher than that in European farm air (Fig. 3B). The five pathogens hosted 43 ARG subtypes, including clinically important beta-lactam (*bla_{OXA}* and *bla_{TEM}*), quinolone (*qnrB* and *qnrS*), and MLS (*erm* and *lnuA*) resistance genes (Fig. 3C). ARG-carrying *Staphylococcus aureus* were detected in the air of all farms, with highest abundance

observed in Chinese pig farms [average 8.5 CPM (copies per million reads), maximum 36.8 CPM] (Fig. 3B). ARG-carrying *Acinetobacter baumannii* and *Campylobacter* spp. were only observed in Chinese chicken farms; the maximum observed abundances were 10.2 and 1 CPM, respectively.

Phylogenetic trees (Fig. 4) were constructed using 59 ARG-carrying metagenome-assembled genomes (MAGs) and 68 isolate genomes from sampled Chinese farms. On average, two ARGs were found per MAG while nine ARGs were identified per isolate genome. The dominant ARG hosts in pig farm air were *Staphylococcus*, *Bacillus*, *Anaerococcus*, *Acinetobacter*, and *Corynebacterium*, accounting for 42% of ARG hosts. In contrast, *Macrocooccus*, *Staphylococcus*, and *Burkholderia* were the main ARG hosts in chicken farm air; they represented 52% of ARG-carrying genera. Many ARG hosts

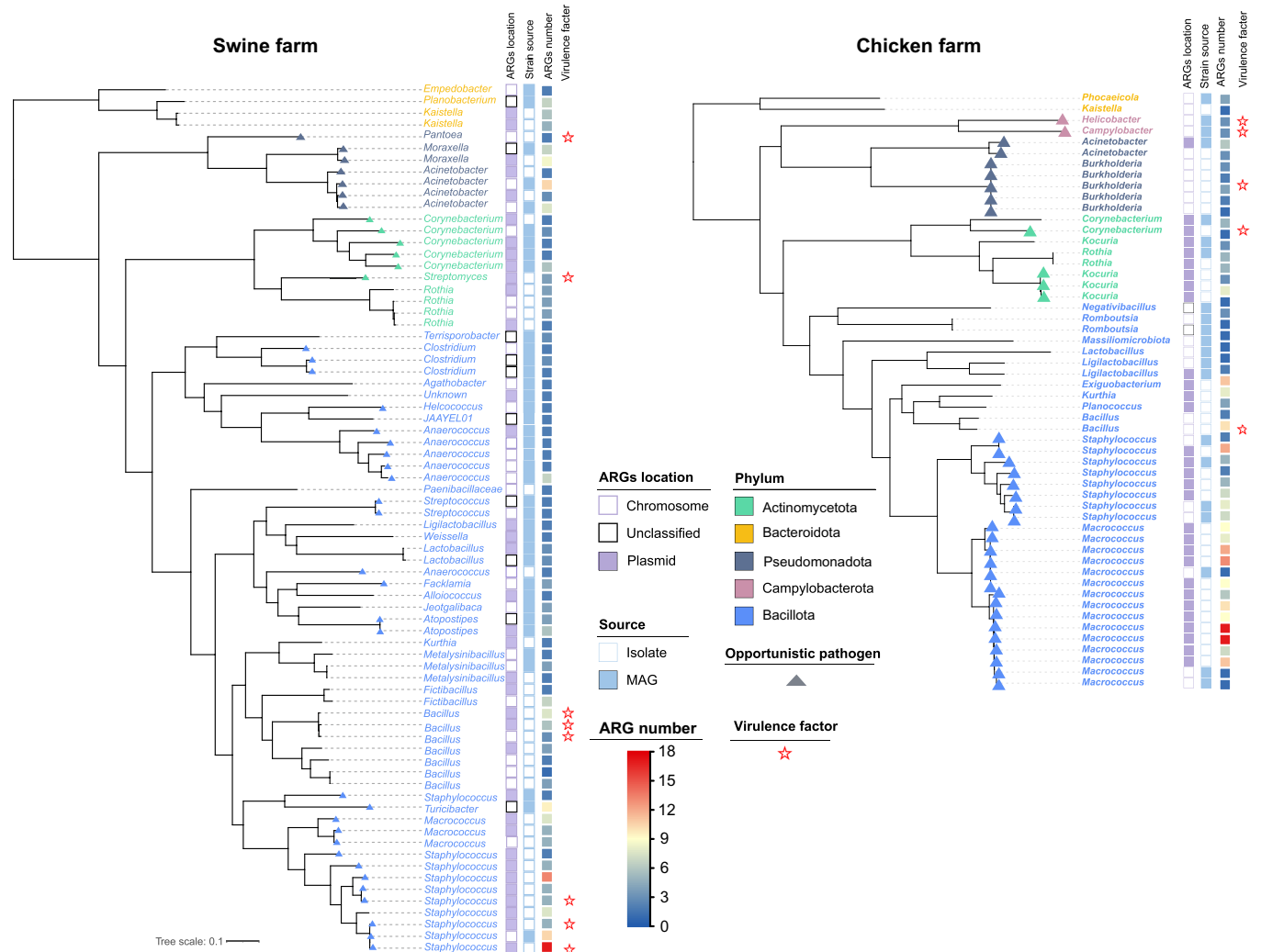


Fig. 4. Phylogenetic tree showing ARGs and virulence factors identified in MAGs and genomes of antibiotic resistant isolates from Chinese swine and chicken farms air.

in livestock farm air were opportunistic pathogens with virulence factors, such as *Staphylococcus*, *Helicobacter*, *Campylobacter*, *Burkholderia*, and *Corynebacterium*.

High Transferability of Air ARGs in Livestock Farms. The genetic location of ARGs in livestock farm air was identified to indicate their transfer potential. In all livestock farm air samples, the abundance of mobile ARGs was significantly ($P < 0.05$) higher than that of immobile ARGs (Fig. 5A). The average abundance of mobile ARGs in the air of chicken and pig farms in China were 124 ± 22.7 CPM and 72 ± 13.2 CPM, which were 22 times and 13 times of that in urban air, respectively (Fig. 5A). The abundance of mobile ARGs in Chinese farm air was over two times of that in European farm air. The European pig farm air had the lowest abundance of mobile ARGs among all livestock farms.

Thirty-three of 68 isolates harbored conjugative plasmids or ICEs carrying ARGs. The three *Acinetobacter* isolates carried conjugative plasmids with ARGs resistant to seven classes of antibiotics such as beta-lactams (*bla*_{TEM} and *bla*_{CARB-2}), MLS (*mphE* and *msrE*), and trimethoprim (*dfrA1* and *dfrA16*). The opportunistic pathogen *Corynebacterium xerosis* had a conjugative plasmid carrying nine ARGs. Another opportunistic pathogen *Staphylococcus haemolyticus* isolated from a pig farm contained 25 ARGs, including 6 ARGs (aminoglycoside, MLS and tetracycline resistance

genes) identified on a conjugative plasmid and 8 ARGs (including 4 beta-lactam resistance genes) on ICEs.

To further assess the transferability of ARGs carried by farm air bacteria to potential pathogens, we determined the conjugation frequency of ARGs in seven antibiotic resistant isolates. We found that two isolates (*Acinetobacter* and *Kaistella* sp002205795) from pig farms successfully transferred their ARGs to *E. ludwigii* GM20. *Acinetobacter* PI-13, carrying 12 mobile ARGs, the conjugative frequency of chlortetracycline and erythromycin resistance genes were 1.58×10^{-3} and 5.25×10^{-5} , respectively (Fig. 5B). *Kaistella* sp002205795 PI-33 harbored seven ARGs, it is conjugative frequency of chlortetracycline and erythromycin resistance genes were 4.02×10^{-4} and 6.94×10^{-5} .

ARG Exposure and Health Risks of Livestock Farm Workers and Near-Farm Residents. The absolute abundance of air resistomes in chicken farms, pig farms, and nearby-farm areas (500 m) were 1.93×10^5 , 1.85×10^6 , and 2.79×10^3 copies per m³, which were 528, 5,045, and 7.6 times of that in the urban air, respectively (SI Appendix, Fig. S6). Chicken and pig farm workers were calculated to have inhaled 0.97 and 9.23 million copies of ARGs per day, equivalent to the cumulative amount of 176 and 1,677 d for the urban residents, respectively (Fig. 6A). The near-farm residents inhaled 41.8 thousand copies of ARGs daily, equivalent

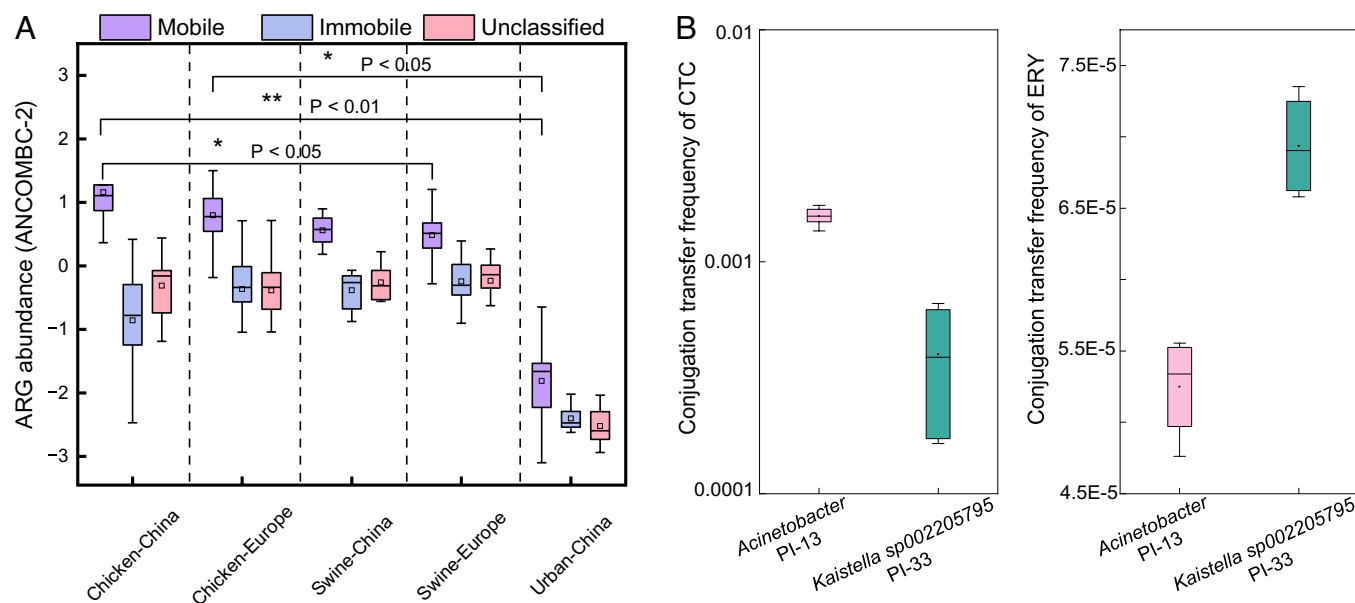


Fig. 5. Mobility of air resistomes in livestock farms. (A) Abundance of mobile and immobile ARGs in different types of air. (B) Conjugation transfer frequency of chlortetracycline (CTC) and erythromycin (ERY) resistance genes from *Acinetobacter* (PI-13) and *Kaistella* sp002205795 (PI-33) to *Enterobacter ludwigii* strain GM20.

to 11.4 d of inhalation for urban residents. The abundance of *intI1* in livestock farms (1,500 copies per m³) was over 130 times higher than that in urban air (SI Appendix, Fig. S7). Two plasmid-mediated colistin resistance genes, *mcr-1* and *mcr-2*, were detected in the air of one chicken farm and two pig farms in China, with highest abundance 132 copies per m³ detected in a pig farm (Fig. 6B). Seven *tetX* variants (*tetX*, *tetX1*–*tetX6*) were found in livestock farm air; the pig farm and chicken farm air had 1,433 and 2,778 copies per m³, respectively (Fig. 6C).

We estimated the resistome risk scores in different types of air samples and in typical ARG reservoirs by MetaCompare (Fig. 6D). The risk score of air resistomes in livestock farms were significantly higher than that for hospital and sewage plant ($P < 0.05$). The risk score of air resistomes in Chinese chicken and pig farms were similar and both were significantly higher than that in European pig farms. Compared to soils and animal feces, livestock farm air had significantly higher risk scores.

Discussion

Livestock farm air is a potential transmission route for antibiotic resistance to humans that has not been fully studied especially for health risk evaluation. We demonstrate that farm air is highly enriched in ARGs. Farm workers inhaled more than 1 million copies of ARGs per day, which is equivalent to the cumulative inhalation of urban residents over several years, highlighting farm air as a high human exposure pathway. We detected two types of emerging ARGs, *mcr* and *tetX*, encoding for resistance to the last-resort drugs colistin and tigecycline, respectively, in the air of Chinese farms. Four plasmid-borne *tetX* variants (*tetX3*–*tetX6*) were detected in all Chinese farms and in 61 of 75 European farms, and the daily exposure of Chinese farm workers was up to 14 thousand copies. Residents living nearby the two sampled farms (500 m) inhaled over 10 times more ARGs per day than urban residents. Studies have shown that ARB and ARGs in farm air can be transmitted up to 10 km away (13, 25), thereby resulting in ARG transmission to humans at a broader scale. The impact of environmental ARG exposure on human resistomes is a key yet debatable question. The ARGs in farm workers was not assessed

in this study, but previous studies have shown that exposure to farms can significantly increase the intestinal resistome abundance of farm workers and veterinary students (16, 26, 27). However, findings are inconsistent across different environments and studies. Exposure to ARGs in coastal water (28) has been linked to an increase of ARG colonization in humans, whereas exposure to air/aerosol ARGs in wastewater treatment plants (WWTP) does not lead to a higher abundance or diversity of ARGs in WWTP workers (29).

Source tracking analysis suggested that the air microbiomes in livestock farms were mainly natural-air sourced, but air resistome component in farms was primarily contributed by animal-associated matrices. Compared to pig farms, a larger proportion of the air resistome in chicken farms was contributed by animal-associated environment, which may be related to the relatively more closed space of chicken farms. Unlike urban air, ARGs diversity and bacterial richness in farm air were not correlated, suggesting that ARGs in farm air are mainly carried by certain hosts (30).

Both culture and culture-independent approaches demonstrate that many ARGs in farm air are highly mobile and carried by pathogens and their close relatives. Combinations of ARG and pathogen posing a major public health threat were observed in farm air, such as *Acinetobacter* and *Staphylococcus* harboring beta-lactam resistance genes (*bla*_{OXA}, *bla*_{TEM}, *bla*_{CARB}, and class A β -lactamase gene). Genomic sequencing of antibiotic resistant isolates presents significant advantage for resolving ARG host, mobility as well as cooccurrence with other ARGs and virulence factors. Isolates can also be used for phenotype identification and conjugation transfer test. However, some ARG hosts may not show resistance phenotypes, or they are difficult to isolate, hence the culture-independent metagenomics is helpful with ARG analyses for such scenarios. We further confirmed the transferability of farm air ARGs into pathogens by conjugation experiments. The recipient *E. ludwigii* belongs to the *Enterobacter cloacae* complex (ECC), which are known to cause a range of life-threatening infections, including respiratory, endocarditis, bacteremia, and urinary tract infections (31). An *Acinetobacter* strain and a *Kaistella* strain isolated from farm air were able to transfer chlortetracycline and erythromycin resistance genes to *Enterobacter* cross order and cross

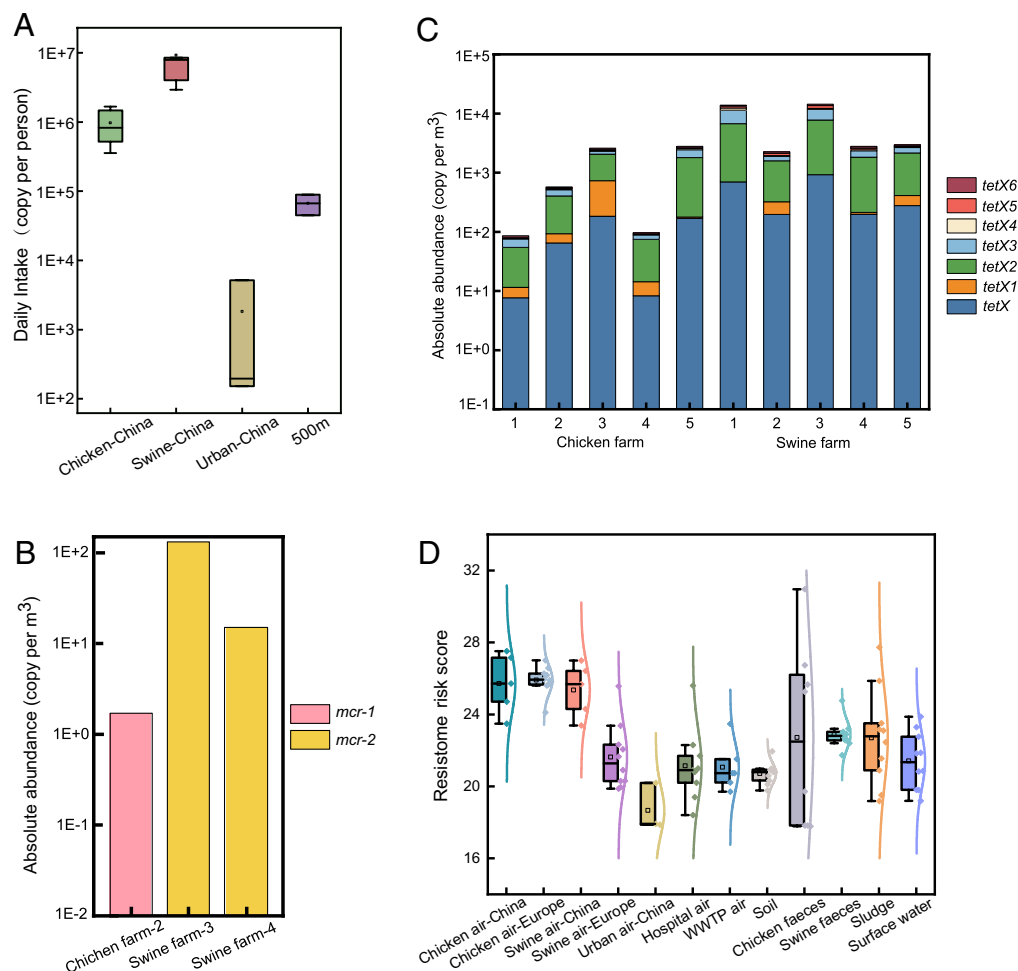


Fig. 6. Exposure and projected risks of air resistomes in livestock farms. (A) Daily inhalation exposure of ARGs by farm workers, residents living near farms, and urban residents. (B) Concentration of *mcr* genes in the air of Chinese farms. (C) Concentration of *tetX* and its variants in the air of Chinese farms. (D) MetaCompare risk score of various ecosystems.

phylum respectively. Fortunately, transfer of *tetX* variants from farm air strains to *Enterobacter* was not observed. Notably, nearly half of our isolated strains harboring conjugative ARG-plasmid and ICEs, indicating a high transfer potential of their ARGs. These results highlight the importance of evaluating resistome risks from multiple dimensions, e.g., isolation and conjugation tests and culture-independent approaches.

We evaluated the resistome risk score in farm air using MetaCompare, which incorporates the relative abundance, mobility, and host pathogenicity of ARGs based on assembled metagenomes (19). The risk score of the air resistomes of livestock farms was significantly higher than those of hospital and wastewater treatment plants, which are two well-recognized ARG hotspots (9, 32). This may be related to the more closed space, higher humidity, and more direct exposure to animals and animal manures in livestock farms (33). Interestingly, although ARGs in farm air were largely derived from animal feces (Fig. 2D) (13), the resistome risk score of farm air was significantly higher than that of animal manure itself, indicating that cross-interface transmission of ARGs from manure to air increases health risk of ARGs. One explanation is that ARG hosts, especially pathogens, have higher ability of self-aggregation and attachment ability to dust particles (34), boosting the resistome risk in the air matrix. Another contributing factor may be the higher survivability of ARB and pathogens in comparison to other fecal bacteria under aerobic environments and exposure to sunlight (35). It is notable

that MetaCompare risk score does not take the absolute ARG exposure into account, which is critical to assess the health risk to humans (12, 19). Given the high exposure of ARGs in farm air, we argue that farm air is a neglected but critical pathway in disseminating antibiotic resistance.

European countries have the longest history in restricting antibiotic use in animal farming, while other countries like China are enforcing similar policies in recent years (23, 24). We used European farm air as a benchmark and compared it with Chinese farms to understand the impacts of antibiotic stewardship on ARG profiles and health risks. We observed significant lower abundance and diversity of air resistomes in European farms, compared to the same type of Chinese farms that we collected. Importantly, the abundance of pathogen-associated and mobile ARGs in European farms were significantly lower than those in Chinese farms, resulting in a significant lower risk score of air resistomes in European farms. To complement and support the observations based on the limited number of farm air samples, we analyzed 1,120 pig fecal metagenomes in EU countries (562) and in China (658) from public databases. Consistent with the observation in farm air resistomes, we found significant higher diversity and abundance in pig manures in Chinese farms than in European farms (SI Appendix, Fig. S8). Notably, even within the nine European countries, a positive correlation between the fecal/air resistome abundance and the antimicrobial usage in livestock farms (36–39). These findings highlight the positive impact of long-term and strict antibiotic use regulations on reducing antibiotic

resistance. Short-term transitions to no-growth-promoting antibiotic policies in China may increase antibiotic use for preventive and therapeutic purposes, contributing to higher resistome levels (24, 39). Also, the legacy effect of ARB/ARGs in farms from before the no-growth-promoting antibiotic regulation can lead to a higher abundance and diversity of antibiotic resistomes in Chinese farms. Other factors, such as farm scale, animal density, and cleaning and disinfection practices, could also impact resistome profiles in farm air (7, 39). However, the significant difference (China vs. European Union) in resistome profile, abundance, and projected health risk is likely due to the different length of time enforcing animal antibiotic use policies. Notably, even with these improvements, the resistome risk score in European farm air remains significantly higher than in urban air, hospital and sewage plant air, animal manures, and soils, highlighting the need for further and sustaining restrictions on antibiotic use and air treatment in animal farming. In summary, our results support that antibiotic stewardship interventions are useful to mitigate antibiotic resistance.

Materials and Methods

TSP Sampling. TSP samples from five pig farms, five chicken farms, and three urban areas were collected during August 2022 in Yangling, China (108.08° N, 34.27° E) (SI Appendix, Table S2 and Fig. S9). To investigate the influence of farm air on surrounding households, an air sample was also collected at a distance of 500 m away from a pig farm and a chicken farm, respectively. High-volume air samplers (LuBo LB-2035, China) with sterile membrane filters were used to collect TSP for 20 h; two replicates were obtained at each site. One membrane filter was stored at −20 °C for genomic DNA extraction, and the other one was stored at +4 °C for ARB isolation within 2 d.

Isolation of Antibiotic Resistant Bacteria. The membrane filter was cut into approximately 1-cm square pieces and added into a 50-mL centrifuge tube containing sterile phosphate buffer solution (pH 7.4). The tubes were shaken at 150 rpm for 1 h followed by centrifugation at 4,000 g for 10 min. Two hundred μ L of the supernatant was spread on Nutrient Broth agar plates containing antibiotics by a spiral inoculator (Interscience, France). The antibiotics and the corresponding concentrations were chlortetracycline 0.5 mg/L, neomycin 1 mg/L, erythromycin 2 mg/L, ciprofloxacin 2 mg/L, and ceftiofur 2 mg/L. The inoculated plates were incubated at 30 °C for 2 to 7 d, then colonies were picked based on morphology and restreaked until pure isolates were obtained. The resistance phenotypes to the above five antibiotics were tested for all isolates. Isolates harboring *tetX* variants were also tested for tigecycline resistance.

Conjugation Experiments. Seven isolates classified to four genera (*Acinetobacter*, *Staphylococcus*, *Macroccoccus*, and *Kaistella*) were selected as donors (SI Appendix, Table S3) since genome sequence analysis suggested that they carry a variety of conjugative ARG-plasmids. A green fluorescence protein labeled *E. ludwigii* strain GM20 with gentamycin resistance was used as the recipient. *Enterobacter* are Gram-negative and commonly found in soil, water, manure, and animal and human-associated environments, and some species are opportunistic pathogens. GM20 used in this study is non-pathogenic but some *E. ludwigii* strains have been reported causing bloodstream and urinary tract infections (40, 41). Donor and recipient cells at a concentration of 10^8 CFU/mL were mixed at 1:1 ratio in 5-mL phosphate buffer solution (pH 7.4). After incubating for 8 h at 30 °C, the mixture was diluted and spread onto nutrient agar plates with gentamycin to enumerate total recipients, and on plates with gentamycin plus another antibiotic to enumerate positive transconjugants. The plates were incubated at 30 °C for 24 h and then colonies of the transconjugants and recipients were counted using a Dark Reader Transilluminator (Clare Chemical Research, USA). The transfer frequency was calculated using the number of transconjugants divided by the total number of recipients.

DNA Extraction and Sequencing. TIAnamp Bacteria DNA kit (TIANGEN, China) and DNeasy Power Soil Pro kit (QIAGEN, Germany) were used for DNA extraction from isolates and membrane filters. DNA purity and concentration were determined by Nanodrop One (Thermo Fisher Scientific, USA) and Qubit 4 fluorometer (Invitrogen, USA). The qualified genomic DNA of air membrane and isolates was

sent to Biomarker Technologies (Beijing, China) for shotgun sequencing on an Illumina NovaSeq PE150 platform. For air membrane DNA, the V3–V4 region of 16S rRNA gene was also sequenced on an Illumina Miseq PE250 sequencer using primers 338F (ACTCTACGGGAGGCAGCA) and 805R (GGACTACHVGGGTWTCTAAT). The air metagenome, whole genome sequences of isolates, and 16S rRNA amplicon data are available through the China National Center for Bioinformation under CRA number CAR015124 (42) and CRA015121 (43), and Zenodo (44) respectively.

Collection of Public Metagenomes and Quality Control. Thirty-five chicken farm air metagenomes and forty pig farm metagenomes from nine European Union countries (Belgium, Bulgaria, Denmark, France, Germany, Italy, the Netherlands, Poland, and Spain) were downloaded from NCBI database (38). Air metagenomes from one chicken farm (22.35° N, 113.49° E), one pig farm (22.19° N, 113.17° E), and six urban sites (22.22° N, 113.55° E) in Zhuhai, China, were also downloaded (45). The details of collected metagenomes are in SI Appendix, Table S4. For all metagenomes, TRIMMOMATIC (v0.39) was used to remove adaptors, primers, and low-quality reads (length < 50 bp or quality score < 20) from raw reads. To eliminate the effect of sequencing depth on resistome and microbial community analyses, 20 million trimmed reads were subsampled from each dataset and used for ARG diversity analyses. Estimation of ARG abundance, assembly, and binning were done using trimmed reads without subsampling.

Microbial Profiling, De Novo Assembly and Binning. Microbial composition in metagenomic data was analyzed using Kraken2 (46) with standard plus database and re-estimated with Bracken (47). The potential human pathogenic bacteria (HPB) were identified according to the priority list defined by World Health Organization (WHO) (48). MEGAHIT (v1.2.9) was used for de novo assembly with default parameters (49). Metagenomic assembled genomes (MAGs) were generated with Metabat2, Maxbin2, and CONCOCT using MetaWRAP (v1.3.2) (50). The output MAGs were refined with bin_refinement module with parameter “-c 50 -x 10.” Isolate genomes were assembled using Spades (v3.15.3) with “--isolate” option. Quality of MAGs and isolate genomes were assessed using CheckM (v1.1.3), and only MAGs with completeness $\geq 50\%$ and contamination $\leq 5\%$ were kept for further analyses. GTDB-Tk (v1.7.0) was used to assign taxonomy to MAGs and isolate genomes.

Identification of ARGs and Their Host Range and Genetic Location. ARG annotation was performed with ARGs-OAP (v2.0) and expressed as copy per cell (51). Only ARGs detected with more than two reads across all samples were retained for further analyses. ARGs and mobile genetic elements (MGEs) on a contig were annotated by BLASTX using SARG (51), ICEberg (52), ISfinder (53), and IntegronFinder (54) databases. The cut-off values for a positive BLASTX hit were identity $\geq 80\%$, alignment length ≥ 25 amino acids, and e-value $\leq 10^{-6}$. ABRicate (v1.0.1) were used to identify virulence factors in MAGs and isolate genomes (55).

Contigs with length over 2 kb were used for identifying host range and genetic location of ARGs. The host of an ARG was identified by classifying ARG-carrying contigs using CAT (v5.0.1) (56). ARG-carrying contigs classified to priority pathogens of antibiotic resistance proposed by WHO (<https://www.who.int/publications/i/item/WHO-EMP-IAU-2017.12>) were further used to calculate pathogen-associated ARG abundance. The abundance of an ARG carried by a contig was calculated by mapping trimmed reads to the contig, and the abundance was expressed as CPM.

The genetic location of an ARG was checked with PlasFlow (v3.5.4) (57) on ARG-carrying contigs. We divided ARGs into three groups to indicate their potential mobility. 1) mobile ARG: the ARG is found on plasmid or flanked by MGEs within 2 kb distance; 2) immobile ARG: the ARG is found on chromosome and no MGE is observed within its 2 kb flanking region; 3) unclassified: the genetic location of the ARG is unclassified, and the ARG is not flanked by MGEs within 2 kb.

Source Tracking of Air Microbiomes and Resistomes in Livestock Farms. To analyze the sources of air microbiomes in livestock farms, we downloaded 16S rRNA gene sequences in 41 ecosystems from the Earth Microbiome Project (58) and used them as potential source environments. The 16S rRNA gene sequences of chicken and swine farm air samples collected in Yangling were used as the sink. All 16S rRNA gene sequences were processed by QIIME2 (v2020.11), and OTUs were generated at 97% similarity. SourceTracker2 (59) was applied for bacterial source tracking with default parameters.

For source tracking of air resistomes in livestock farms, 117 metagenomes from six ecosystems (soil, chicken manure, pig manure, chicken intestine, pig intestine, and human) were collected from NCBI and ENA databases (SI Appendix, Table S5). The 117 metagenomes constituted the putative sources and their resistome profiles were analyzed using ARGs-OAP (v2.0). Source Tracker2 was used based on resistome profiles in the source and sink, to estimate the relative contribution of various environments to air resistomes in livestock farms.

Risk Evaluation of Air Resistomes and Daily ARG Inhalation Exposure.

To compare the risks between air resistome in livestock farms and in typical ARG hotspots, we downloaded metagenomes of hospital air (32), wastewater treatment plant air (60), soils, chicken feces, pig feces, sludge, and surface water (SI Appendix, Table S5). Resistome risk score in each environment was calculated based on the abundance and mobility of antibiotic resistomes as well as ARG hosts' pathogenicity using MetaCompare (19).

The total bacterial population (expressed as copy per m³) in farm air and in near-farm (500 m) air samples was determined using a real-time qPCR platform (BioRad CFX Connect, USA) with 16S rRNA gene primers 341F (CCTACGGGAGGCGAG) and 806R (ATTACCGGCTGCTGG). The specific reaction system and qPCR conditions were described by Sun et al. (16). The absolute abundance of an ARG was calculated by using relative abundance (estimated with metagenomic data, normalized by 16S copy number) of the ARG multiplied by the absolute abundance of 16S rRNA gene in the same sample. The daily inhalation of ARGs for farm workers (8 h on farm and 16 h in urban area per day), near-farm residents (24 h in near-farm area per day), and urban residents (24 h in urban area per day) was estimated based on an average air inhalation rate of 625 L air per hour (32). The absolute abundance of *int1* in farm air samples was quantified using *int1*-F (CTGGATTTCGATCAGGCACG) and *int1*-R (ACATGCGTGTAATCATCTGCG).

Statistical Analysis. The microbial and ARG abundance matrices were subjected to zero replacement using the "SQ" method using the *cmultRpl* function in the *zComposites* R software package (version 1.3.4) (61). The additive log-ratio (ALR) of ARG abundance for each sample is obtained by calculating the logarithmic ratio of ARG-assigned fragment matrices. The abundance of an ARG carried by a contig was converted through centered log-ratio (CLR) (R software package of *Compositions* version 2.0.8). Differential abundance analysis was

based on ANCOMBC-2 with ANCOMBC (v2.8.1) R package. Spearman's correlation coefficients were calculated to assess the relationship between bacterial diversity and ARG diversity. Principal components analysis (PCA) based on ALR coefficient matrix was used to estimate the variation in resistome and bacterial profiles among samples. The correlation between resistome profile and bacterial profile was evaluated using Procrustes analysis with 9,999 permutations. The phylogenetic tree of isolate genomes and MAGs was inferred using GTDB-tk based on a set of 120 universal single-copy marker genes in bacteria. The tree was visualized using Interactive Tree Of Life (<https://itol.embl.de>). PCA, Procrustes analysis, Spearman correlation analysis and heatmap illustration were done with R software (v4.1.3).

Data, Materials, and Software Availability. The sequence data used in this study are available at <https://ngdc.cncb.ac.cn/gsa/browse/CRA015124> (42), <https://ngdc.cncb.ac.cn/gsa/browse/CRA015121> (43) and <https://zenodo.org/records/15183489> (44). All other study data are included in the article and/or supporting information.

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Author affiliations: ^aInterdisciplinary Research Center for Soil Microbial Ecology and Land Sustainable Productivity in Dry Areas, Northwest A&F University, Shaanxi 712100, China; ^bDepartment of Environmental Science, College of Natural Resources and Environment, Northwest A&F University, Shaanxi 712100, China; ^cKey Laboratory of Plant Nutrition and the Agri-environment in Northwest China, Ministry of Agriculture, Northwest A&F University, Shaanxi 712100, China; ^dDepartment of Infectious Diseases, Second Affiliated Hospital of Xi'an Jiaotong University, Shaanxi 710004, China; ^eDepartment of Microbiology, College of Life Sciences, Northwest A&F University, Shaanxi 712100, China; and ^fCenter for Microbial Ecology, Michigan State University, East Lansing, MI 48824

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